CHAPTER 06

Profit, Loss and Discount

This chapter is one of the important chapters in the Quantitative Aptitude section. Since calculating profit, loss and discount is our daily need, specially, if we are into business. Even calculating profit and loss is important for everyone who makes any buying or selling decisions of any sort of goods and services, because of one's propensity to earn more, spend less, increase profit and reduce loss. If people have multiple options to buy and sell they compare the prices and try to take the best decision. That's why there is hardly any competitive exam which does not include the problems based on this chapter. In fact, we can find a huge number of problems in Data Interpretation section.

6.1 Theory and Concepts

In day-to-day life we sell and purchase the things as per our requirement. A customer can get things in the following manner:

Manufacturer (or producer) → Whole-saler (dealer) → (Shopkeeper) Retailer (or sales person) → Customer

Terminology

Cost price (CP) The money paid by the shopkeeper to the manufacturer or whole-saler to buy the goods is called the cost price (CP) of the goods purchased by the shopkeeper.

NOTE If an article is purchased for some amount and there are some additional expenses on transportation labour, commission etc., these are to be added in the cost price. Such expenses are called overhead expenses or overheads.

Selling Price (SP) : The price at which the shopkeeper sells the goods is called the selling price (SP) of the goods sold by the shopkeeper.

Profit : If the selling price of an article is more than its cost price, then the dealer (or shopkeeper) makes a profit (or gain) *i.e.,* Profit = SP – CP; SP > CP

Loss : If the selling price of an article is less than its cost price, then the dealer suffers a loss.

i.e., loss = CP – SP; CP > SP

Chapter Checklist

- *Theory and Concepts*
- *Mark up and Discount*
- *Common Gain or Loss Concept*
- *Selling Price and Cost Price Equality*
- *Profit-Loss Factors*
- ***CAT Test***

Important Formulae

(i) Profit $= SP - CP$ (ii) Loss $= CP - SP$

(iii) Profit percentage $= \dfrac{\text{profit}}{\text{cost price}} \times 100$

(iv) Loss percentage $= \dfrac{\text{loss}}{\text{cost price}} \times 100$

(v) $SP = \left(\dfrac{100 + \text{gain}\%}{100} \times CP\right) = \left(\dfrac{100 - \text{loss}\%}{100} \times CP\right)$

(vi) $CP = \left(\dfrac{100}{100 + \text{gain}\%} \times SP\right) = \left(\dfrac{100}{100 - \text{loss}\%} \times SP\right)$

(vii) $SP = (100 + k)\%$ of CP; when profit $= k\%$ of CP

(viii) $SP = (100 - k)\%$ of CP; when loss $= k\%$ of CP

NOTE Profit or loss is always calculated on the basis of cost price unless otherwise mentioned in the problem.

***Exp.* 1)** A fruit seller buys 300 oranges at 5 for ₹ 8 and sold at 2 for ₹ 5. Find :

(i) the cost price of each orange
(ii) the selling price of each orange
(iii) profit or loss on selling one orange
(iv) his total profit or loss on selling all the oranges.

Solution

(i) Since, the cost price of 5 oranges = ₹ 8

$\therefore$ Cost price of one (or each) orange = $₹ \dfrac{8}{5} =$ ₹ 1.60

(ii) Since, the selling price of 2 oranges = ₹ 5

$\therefore$ The selling price of one orange = $₹ \dfrac{5}{2} =$ ₹ 2.50

(iii) Since, SP is more than CP, there is a profit.
So profit on selling one orange = SP – CP
= 2.5 – 1.6 = ₹ 0.90

(iv) Profit on selling all the oranges
= ₹ (0.90 × 300) = ₹ 270
Hence, the total profit on selling all oranges = ₹ 270

***Exp.* 2)** A shopkeeper buys 100 eggs at ₹ 1.20 per piece. Unfortunately 4 eggs got spoiled during transportation. The shopkeeper sells the remaining eggs at ₹ 15 a dozen. Find his profit or loss.

Solution Cost price of all eggs = ₹ 100 × 1.2 = ₹ 120

Selling price of one egg $= \dfrac{15}{12} =$ ₹ 1.25

$\therefore$ Selling price of 96 eggs $= 96 \times \dfrac{15}{12} =$ ₹ 120

Hint After spoiling 4 eggs, only 96 eggs are left.
Now, since the total selling price and total cost price is same, the shopkeeper neither makes a profit nor suffers a loss.

***Exp.* 3)** Aviral purchased a computer for ₹ 47,000. He had to sell it for ₹ 45,800. Find his profit or loss per cent.

Solution Since, SP < CP, there will be loss

$\text{loss}(\%) = \dfrac{\text{loss}}{CP} \times 100 = \dfrac{47000 - 45800}{47000} \times 100 = \dfrac{1200}{47000} \times 100 = 2\dfrac{26}{47}\%$

***Exp.* 4)** A dealer sold 600 quintals of sugar at a profit of 7%. If a quintal of sugar cost him ₹ 1600, find his total profit and the selling price.

Solution CP = 1600 × 600 = ₹ 9,60,000

Rate of Profit = 7%

$\text{Profit} = 9,60,000 \times \dfrac{7}{100} =$ ₹ 67200

$\therefore$ SP = CP + Profit = ₹ 9,60,000 + ₹ 67,200 = ₹ 1027200

Thus, profit = ₹ 67200 and selling price = ₹ 10,27,200

***Exp.* 5)** A dealer buys 200 quintals of wheat at ₹ 1200 a quintal. He spends ₹ 10,000 on transportation and storage. Then, he sells the wheat at ₹ 13 per kg. Find his profit or loss. Also calculate it as a percentage.

Solution CP = 1200 × 200 = ₹ 2,40,000

Transportation and storage cost = ₹ 10,000

Total CP = 2,40,000 + 10,000
= ₹ 2,50,000

Total SP = 13 × 200 × 100 = ₹ 2,60,000

$\therefore$ Now, since SP > CP, hence there will be profit.

Profit = SP – CP = ₹ (2,60,000 – 2,50,000) = ₹ 10,000

$\text{Profit}(\%) = \dfrac{\text{Profit}}{CP} \times 100 = \dfrac{10,000}{2,50,000} \times 100 = 4\%$

***Exp.* 6)** Find the cost price of an article which is sold for ₹ 220 at a loss of 12%.

Solution SP = ₹ 220, Loss = 12%

Let CP = ₹ x

Then, SP = 88% of CP

$220 = \dfrac{88}{100} \times x \Rightarrow x = 250$

Therefore, cost price = ₹ 250

***Exp.* 7)** By selling a colour TV for ₹ 23520, a dealer suffers a loss of 4%. What is the cost price of the colour TV? At what price should he sell it to gain 8%?

Solution SP = 96% of CP $\because$ (loss is 4%)

$\therefore$ $23520 = \dfrac{96}{100} \times CP$

CP = 24500

Now, gain % = 8%

$\therefore$ New selling price = (100 + 8)% of CP

$= \dfrac{108}{100} \times 24500 =$ ₹ 26460

Introductory Exercise 6.1

1. A towel is sold for ₹ 198 at a gain of 10%. What is the cost price of the towel? At what price must it be sold to gain 25%?
2. A man sold a watch at ₹ 6000, at a loss of $33\frac{1}{3}\%$. Find the cost price?
3. By selling a shirt for ₹ 285 a shopkeeper loses 5%. At what price should he sell the shirt, so as to gain 15%?
4. Sufyan bought 1200 eggs at ₹ 16 a dozen. At what price per hundred must he sell the eggs, so as to earn a profit of 15%?
5. Ram Singh purchased two camels for ₹ 18,000 and ₹ 15,000 respectively. He sold them at a loss of 15% and a gain of 19% respectively. Find the selling price of each of the camels. Also find the overall gain or loss per cent in the transaction.

6.2 Markup and Discount

Marked price: Basically to avoid loss due to bargaining by the customer and to get the profit over the cost price trader increases the cost price by a certain value, this increase in value over cost price is known as markup and the increased price (*i.e.,* CP + Markup) is called the marked price or printed price or list price of the goods.

Marked price = CP + Markup

Marked price = CP + (% markup on CP)

Generally goods are sold at marked price, if there is no further discount, then in this case selling price equals to marked price.

Discount : Discount means reduction of marked price to sell at a lower rate or literally discount means concession. Basically it is calculated on the basis of marked price.

$\therefore$ Selling price = Marked price – Discount

or Selling price = Marked price (MP)

– (% discount on MP)

Since, marked price = CP + % markup on CP

Remember **markup** is calculated on the basis of CP while **discount** is calculated on the basis of MP.

In general, CP < SP < MP at profit

CP = SP < MP at no profit no loss

SP < CP < MP at loss

Also, CP $\xrightarrow{\text{Markup}}$ MP

and SP $\xrightarrow{\text{discount}}$ MP

NOTE You will find wide application of percentage calculation, percentage change graphic etc. So, to move further you must know the concepts of percentages.

***Exp.* 8)** If the cost price of an article is ₹ 300 and the per cent markup is 20%. What is the marked price?

Solution MP = CP + % markup on CP

$$= 300 + 300 \times \frac{20}{100}$$

MP = ₹ 360

Alternatively $SP = 300 \times 1.2 = 360$

***Exp.* 9)** If the marked price of an article is ₹ 450 and markup percentage is 12.5%, what is the cost price?

Solution MP = 112.5 of CP

$$\therefore \quad 450 = \frac{1125}{100} \times CP$$

$$450 = \frac{9}{8} \times CP$$

$$\Rightarrow \quad CP = ₹\ 400$$

Alternatively Change your outlook to visulize the problem differently for smarter calculation.

You can see that MP is 12.5% (*i.e.,* 1/8 times) greater than CP. So, CP will be 1/9 times less than MP. This percentage (or fraction) change rule has been throughly discussed through different illustrations in percentage chapter. In my opinion if you change your observation *i.e.,* you mould your thinking in a CAT oriented approach you will not change only your attitude and aptitude but the whole scenario will be changed. So, look this problem like me :

CP $\xrightarrow{1/8 \uparrow}$ MP $\left(\frac{n}{d+n}\right)$ ($n \rightarrow$ Numerator, $d \rightarrow$ denominator)

MP $\xrightarrow{1/9 \downarrow}$ CP

So, the CP will be $\frac{1}{9}$ times less than MP, which is rather too much easier to calculate than the traditional method given at the beginning of the solution.

***Exp.* 10)** If the marked price of an article is ₹ 660 and the discount is 10%, then what is the selling price of the article?

Solution **General Solution**

$$SP = MP - Discount$$

$$SP = 660 - 660 \times \frac{10}{100} = 594$$

Alternatively SP = 90% of MP

$$SP = 0.9 \times 660 = 594$$

Alternatively We can see that when SP is 10% (*i.e.*, $\frac{1}{10}$) less than MP, it means MP is $\frac{1}{9}$ times greater than SP.

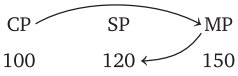
Therefore, $MP = \frac{10}{9} SP$ $\left(1 + \frac{1}{9} = \frac{10}{9}\right)$

Ultimately I would like to say that you should see the core of the problem and solution both, instead of taking help from any formula. In other words formulae make you logically blind particularly in context of CAT where your intelligence is tested rather than your memory and theory of the maths.

***Exp.* 11)** If the markup percentage of an article is 50% and discount percentage is also 20%, then the profit percentage will be :

(a) 10% (b) 0% (c) 30% (d) 20%

Solution See the smartest calculation technique and follow me to be smarter.



Therefore, profit = 20% $\left(\frac{120 - 100}{100} \times 100 = 20\%\right)$

Explanation: MP = 150% of CP (∵ markup is 50%)

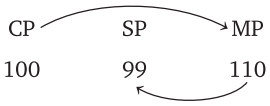
SP = 80% of MP (∵ discount is 20%)

Once again I am reiterating the gist of this chapter. What is that gist? The gist is that this chapter test nothing but your grip on the concepts of percentage change.

I think CP, SP, MP etc., are just the words, the crux is that how CP, SP and MP etc., change with respect to each other. So those students who have grasped and internalised the most important concept of percentage change, they will not find any difficulty in understanding the concept of the profit-loss problems. Believe me when your visualisation will be improved, you will be master on this chapter. So try to have a different approach.

***Exp.* 12)** A trader markup the goods by 10% and then gives a discount of 10%. What is the profit or loss percentage?

Solution



So, there is a loss of 1%.

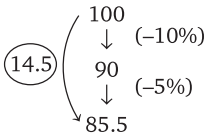
Explanation: MP = 110% of CP (10% markup)

SP = 90% of MP (10% discount)

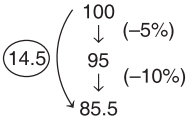
***Exp.* 13)** Successive discount of 10% and 5% is equivalent to

(a) 16.5% (b) 15% (c) 15.5% (d) 14.5%

Solution



NOTE There is no difference in two different cases *i.e.*, either you decrease 100 at first by 10% and then by 5% or decrease 100 at first by 5% and then by 10%.



Explanation : $100 \times 0.9 \times 0.95 = 85.5$

∴ Single discount equivalent to 10% and 5% is 14.5%

$(100 - 85.5)\%$

or $(100 \times 0.95 \times 0.9 = 85.5)$

6.3 Common Gain or Loss Concept

When two articles are sold at the same price, but one article at a profit and the other at a loss and the percentage profit is the same as the percentage loss. In this case there is always a loss.

$$\text{loss (\%)} = \left(\frac{\text{Common gain or loss}}{10}\right)^2$$

***Exp.* 14)** A man sells two wrist watches one at a profit of 10% and another at a loss of 10%, but the selling price of each watch is ₹ 200. Find the :

(1) percentage profit or loss (2) net amount of profit or loss

Solution

(1) Since, there is always loss. (**Logic:** To get the logic you should refer the percentage chapter)

Now, $\text{loss \%} = \left(\frac{\text{Common gain or loss}}{10}\right)^2 \%$

$$\text{loss \%} = \left(\frac{10}{10}\right)^2 \% = 1\%$$

(2) Amount of loss.

Total SP = ₹ 400 (200 + 200)

Again, total SP = 99% of CP (1% loss)

$$400 = \frac{99}{100} \times CP$$

$\Rightarrow$ CP = ₹ 404.04

$\therefore$ loss = CP – SP

= ₹ 4.04 (= 404.04 – 400)

***Exp.* 15)** There were two articles and the sum of cost prices of these articles is ₹ 500. One of them was sold at a profit of 20% and another at a loss of 20%. Besides if the selling prices of both the articles were same. Find the amount of overall loss.

Solution $x \times \frac{120}{100} = (500 - x)\frac{80}{100} \Rightarrow x = 200$

So, the CP of profit yielding article = ₹ 200

and the CP of loss giving article = ₹ 300 (500 – 200)

and the common SP = ₹ 240 = (200 × 1.2 = 300 × 0.8)

So the loss = CP – SP = 500 – 2 × 240 = ₹ 20

6.4 Selling Price and Cost Price Equality

When the total selling price of '*m*' articles is same as the total cost price of '*n*' articles, where each article is similar

***Exp.* 16)** If the cost price of 15 apples is same as the selling price of 20 apples. What is the gain or loss per cent?

Solution CP of 15 apples = SP of 20 apples

$\Rightarrow \quad CP \times 15 = SP \times 20 \Rightarrow \frac{CP}{SP} = \frac{4}{3}$

So you can see that CP > SP, therefore, there will be loss.

Now consider CP = 4, then SP = 3

$\therefore$ loss = 1

$\therefore \quad \text{loss (\%)} = \frac{\text{loss}}{CP} \times 100 = \frac{1}{4} \times 100 = 25\%$

loss = 25%

***Exp.* 17)** If the selling price of 10 CDs is the same as the cost price of 12 CDs. What is the profit or loss per cent?

Solution SP of 10 CDs = CP of 12 CDs

$\Rightarrow \quad SP \times 10 = CP \times 12 \Rightarrow \frac{SP}{CP} = \frac{12}{10} = \frac{6}{5}$

$\Rightarrow$ SP > CP, therefore there will be a profit

$$\text{Profit (\%)} = \frac{(SP - CP)}{CP} \times 100 = \frac{(6-5)}{5} \times 100 = 20\% \text{ (profit)}$$

When the articles are sold at the cost price, but the quantity bought and sold are distinct

When a person recovers the cost price of 'm' articles by selling '*n*' articles ($n < m$), then

$$\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{m-n}{n} \times 100$$

In this case money is equated in terms of number of (or amount) articles. For your convenience you can try assuming the CP of an article as ₹ 1. (or ₹ 100)

***Exp.* 18)** A dealer by selling 12 oranges gets the cost price of 15 oranges. What is the percentage profit?

Solution $\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{15-12}{12} \times 100 = 25\%$

Alternatively Suppose CP of one orange = ₹ 1

then CP of 12 oranges = ₹ 12

and SP of 12 oranges = CP of 15 oranges = ₹ 15

$\therefore \quad \text{profit} = \frac{15-12}{12} \times 100 = 25\%$

***Exp.* 19)** By selling 8 bananas, a fruit seller gains the selling price of 1 banana. Calculate his gain per cent.

Solution Let the SP of one banana = ₹ 1,

Then SP of 8 bananas = ₹ 8

and profit = Re. 1

$\therefore \quad CP = 8 - 1 = ₹ 7$

$\therefore \quad \text{Profit \%} = \frac{1}{7} \times 100 = 14\frac{2}{7}\%$

***Exp.* 20)** By selling 18 chocolates, a vendor loses the selling price of 2 chocolates. Find his loss per cent.

Solution Let the SP of 1 chocolate = ₹ 1

$\therefore \quad$ SP of 18 chocolate = ₹ 18 and loss = ₹ 2

$\therefore \quad CP = SP + \text{loss} = 18 + 2 = ₹ 20$

$\therefore \quad \text{Percentage loss} = \frac{\text{loss}}{CP} \times 100 = \frac{2}{20} \times 100 = 10\%$

***Exp.* 21)** A trader sell all his articles at the cost price but gives 10% less amount as he should give. What is his percentage profit?

Solution $\text{Profit (\%)} = \frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{10}{90} \times 100 = 11\frac{1}{9}\%$

Since, if we assume that the CP of 1 article is ₹ 1.

Now, since he gives only 90% article instead of 100% and save 10% article. So, his profit will be the equal to the remaining articles (over the sold articles). It means when he sells the articles (actually) worth ₹ 90, then he gains by articles worth ₹ 10.

Hence, profit % $= \frac{10}{90} \times 100 = 11\frac{1}{9}\%$

(The selling price $= \frac{100}{90} = \frac{10}{9} = ₹ 1.11$)

***Exp.* 22)** A trader by means of his false balance defrauds to the extent of 10% in buying goods and also defrauds to 10% in selling. Find his gain per cent.

Solution Let the actual CP of an article be ₹ 1, then the

effective CP $= \frac{100}{110} = \frac{10}{11}$

(Since, he purchases 110 articles by paying ₹ 100)

Again, SP $= \frac{100}{90} = \frac{10}{9}$

(Since, he sells only 90 articles charging the CP of 100 articles)

$\therefore \quad \text{Gain \%} = \frac{SP - CP}{CP} \times 100 = \frac{\frac{10}{9} - \frac{10}{11}}{\frac{10}{11}} \times 100$

$= \frac{20}{99} \times \frac{11}{10} \times 100 = \frac{200}{9} = 22\frac{2}{9}\%$

Introductory *Exercise 6.2*

1. The cost price of a scooter is ₹ 20,000 and the profit per cent is 12%. What is the selling price?
(a) 2400 (b) 22040
(c) 2600 (d) 22400

2. The SP of an article is ₹ 3200 and the profit per cent is $33\frac{1}{3}\%$. Find the cost price.
(a) ₹ 20000 (b) ₹ 2000
(c) ₹ 2400 (d) ₹ 3000

3. The CP of an article is 5/6th of the SP. What is the percentage profit or loss?
(a) 20% loss (b) 16.66% profit
(c) 16.66% loss (d) 20% profit

4. The MP of a camera is 3/2 of the CP and SP is 9/10 of MP. Find the percentage profit or loss.
(a) 25% profit (b) 35% profit
(c) 33.33% loss (d) none of these

5. The MP of an article is 30% higher than its CP and 20% discount is allowed on this article, then the profit percentage :
(a) 10% (b) 14% (c) 4% (d) 26%

6. An article is sold for ₹ 1980 at 10% profit. What is the cost price?
(a) ₹ 198 (b) ₹ 1800
(c) ₹ 1900 (d) ₹ 1600

7. On selling an article for ₹ 576 a trader loses 4%. In order to gain $4\frac{1}{6}\%$, he must sell that article for :
(a) ₹ 636 (b) ₹ 676
(c) ₹ 625 (d) can't be determined

8. The per cent profit made when an article is sold for ₹ 56 is thrice as when it is sold for ₹ 42. The cost price of the article is :
(a) ₹ 48 (b) ₹ 49
(c) ₹ 50 (d) ₹ 35

9. A shopkeeper uses a weight of 460 g instead of 500 g and sells the articles at the cost price. What is the profit percentage?
(a) 40% (b) 23%
(c) $8\frac{16}{23}\%$ (d) 20%

10. A trader uses a weight of 920 g instead of 1 kg and sells the articles at the marked price which is 15% above the cost price. Find the profit percentage.
(a) 20% (b) 23%
(c) 25% (d) can't be determined

11. If a gift pack is sold at a gain of 6% instead of at a loss of 6%, then the seller gets ₹ 6 more. The cost price of the article is :
(a) ₹ 60 (b) ₹ 66
(c) ₹ 50 (d) ₹ 36

12. A man sells a bicycle at a gain of 10%. If he had bought it at 10% less and sold it for ₹ 132 less, he would have still gained 10%. The cost price of the article is :
(a) ₹ 1000 (b) ₹ 1200
(c) ₹ 1500 (d) ₹ 1320

13. An item costing ₹ 600 is being sold at 20% loss. If the price is further reduced by 12.5%, the selling price will be :
(a) ₹ 400 (b) ₹ 380
(c) ₹ 420 (d) ₹ 525

14. While selling an electric fan, a dealer gives a discount of 5%. If he gives a discount of 8%, he earns ₹ 36 less as profit. The marked price of the fan is :
(a) ₹ 1000 (b) ₹ 1200
(c) ₹ 800 (d) none of these

15. If a commission of 10% is given on the marked price of a book, the publisher gains 20%. If the commission is increased to 15%, the gain of publisher is :
(a) 13.33% (b) 15%
(c) 18% (d) data insufficient

16. A retailer buys a cellphone at a discount of 15% and sells it for ₹ 5865. Thus, he makes a profit of 15%. The discount is :
(a) ₹ 200 (b) ₹ 850
(c) ₹ 750 (d) ₹ 900

17. At what per cent above the cost price must an articles be marked, so as to gain 17% after allowing a discount of 10%?
(a) 34% (b) 70% (c) 30% (d) 27%

18. A merchant marks his goods at ₹ 900 and allows a discount of 25%. If he still gain 12.5%, then the cost price of the article is :
(a) ₹ 500 (b) ₹ 600
(c) ₹ 720 (d) can't be determined

19. A vendor buys oranges @ ₹ 2 for 3 oranges and sells them at a rupee each. To make a profit of ₹ 10, he must sell :
(a) 10 oranges (b) 20 oranges
(c) 30 oranges (d) 40 oranges

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 Abhishek and Bhanu both are dealers of KML scooters. The price of a KML Scooter is ₹ 28,000. Abhishek gives a discount of 10% on whole, while Bhanu gives a discount of 12% on the first ₹ 20,000 and 8% on the rest ₹ 8000. What is the difference between their selling prices?
(a) ₹ 240 (b) ₹ 420
(c) ₹ 640 (d) none of these

2 A trader sells two articles, one at a loss of 10% and another at a profit of 15% but finally there is no loss or gain. If the total sale price of these two articles is ₹ 30,000, find the difference between their cost prices :
(a) ₹ 5000 (b) ₹ 6000
(c) ₹ 7500 (d) none of these

3 A milkman purchases the milk at ₹ x per litre and sells it at ₹ $2x$ per litre still he mixes 2 litres water with every 6 litres of pure milk. What is the profit percentage?
(a) 116% (b) 166.66%
(c) 60% (d) 100%

4 60% goods are sold at 5% loss while rest are sold at 10% profit. If there is a total profit of ₹ 100, then the worth of goods sold is :
(a) ₹ 6000 (b) ₹ 5000
(c) ₹ 10000 (d) none of these

5 A retailer bought 20 kg tea at a discount of 10%. Besides 1 kg tea was freely offered to him by the wholesaler at the purchase of 20 kg tea. Now, he sells all the tea at the marked price to a customer. What is profit percentage of retailer?
(a) 30% (b) 12%
(c) 16.66% (d) none of these

6 Two articles are sold at the same price. One at a profit of 75% and another one at a loss of 30%. What is the overall profit or loss?
(a) 22.5% profit (b) 57.5% profit
(c) $13\frac{2}{7}$% loss (d) none of these

7 What is percentage profit in selling an article at a discount of 20% which was earlier being sold at a 40% profit?
(a) 20% (b) 14%
(c) 28% (d) 12%

8 A man bought 18 oranges for a rupee and sold them at 12 oranges for a rupee. What is the profit percentage?
(a) 33.33% (b) 50%
(c) 66.66% (d) none of these

9 A dealer buys a product at ₹ 1920, he sells at a discount of 20% still he gets the profit of 20%. What is the selling price of that product?
(a) ₹ 2304 (b) ₹ 1536
(c) ₹ 2200 (d) it is not possible

10 Tinkawala purchased the articles for ₹ 123684. He sold 60% of those at a profit of 16.66% and rest at a loss. Find the loss percentage on the remaining if the overall loss is 14%?
(a) 20% (b) 30% (c) 60% (d) 66.66%

11 What should be the minimum markup percentage such that after giving a discount of $66\frac{2}{3}$%, there will not be a loss?
(a) 200% (b) 133.33%
(c) 100% (d) 150%

12 The ratio of cost price and marked price of an article is 2 : 3 and ratio of percentage profit and percentage discount is 3 : 2. What is the discount percentage?
(a) 16.66% (b) 20% (c) 25% (d) 33.33%

13 A dealer gives as much discount (in per cent) as the markup (in per cent) above the cost price. What is the profit or loss per cent?
(a) 10% (b) 1%
(c) 4% (d) Can't be determined

14 In the above problem, what is the markup percentage when there is a loss in percentage be half of the percentage markup?
(a) 20% (b) 40%
(c) Can't be determined (d) none of these

15 A shopkeeper sold 12 cameras at a profit of 20% and 8 cameras at a profit of 10%. If he had sold all the 20 cameras at a profit of 15%, then his profit would have been reduced by ₹ 36. What is the cost price of each camera?
(a) 100 (b) 150 (c) 180 (d) 220

16 Mr. Mittal purchased a car for ₹ 3,00,000 and a bike for his son for ₹ 1,00,000. He sold the car at a profit of 10% and bike at a loss of 20%. What is the net gain or loss?
(a) 2% gain (b) 1.5% loss
(c) 2.5% loss (d) 2.5% gain

17 A trader sells 20 kg of sugar at ₹ 400. A customer asks 20% discount and he agrees to it but instead of 1 kg he gives 4% less sugar. What is the effective discount that the customer gets?
(a) 16% (b) 16.66%
(c) 15.5% (d) 19.6%

18 The profit percentage on the three articles A, B and C is 10%, 20% and 25% and the ratio of the cost price is 1 : 2 : 4. Also the ratio of number of articles sold of A, B and C is 2 : 5 : 2, then the overall profit percentage is :
(a) 18.5% (b) 21%
(c) 75% (d) none of these

19 The marked price of an article is increased by 25% and the selling price is increased by 16.66%, then the amount of profit doubles. If the original marked price be ₹ 400 which is greater than the corresponding cost price by 33.33%, what is the increased selling price?
(a) 240 (b) 360
(c) 420 (d) 600

20 A shopkeeper calculated his profit per cent on the selling price which comes out to be 30%. If it had been calculated as usual on the cost price, then what is the required percentage profit?
(a) $42\frac{6}{7}\%$ (b) $4\frac{7}{27}\%$
(c) $7\frac{6}{42}\%$ (d) none of these

21 An item was sold after giving two successive discount of 20% and 10% respectively. If the item was sold for ₹ 468. The marked price of that item is :
(a) ₹ 600 (b) ₹ 500
(c) ₹ 575 (d) ₹ 650

22 The cost price of an article 'A' is ₹ 160 and selling price of another article 'B' is ₹ 240. If the selling price of A will be equal to the cost price of B, then the profit after selling A is 20%. What is the profit on 'B'?
(a) 16.66% (b) 50%
(c) 25% (d) none of these

23 A single discount equivalent to three successive discounts of 5%, 10%, 20% is :
(a) 68.4% (b) 35%
(c) 31.6% (d) 32%

24 Ragini purchases oranges at ₹ 10 per dozen and sells them at ₹ 12 for every 10 oranges. What is the profit percentage?
(a) 40% (b) 44%
(c) 60% (d) 48%

25 A trader uses a weighing balance that shows 1250 g for a kilogram. He further marks-up his cost price by 20%. What is the profit percentage?
(a) 5% (b) 45%
(c) 50% (d) 30%

26 On selling an article for ₹ 240, a trader loses 4%. In order to gain 10% he must sell that article for :
(a) ₹ 275 (b) ₹ 340 (c) ₹ 320 (d) ₹ 264

27 A merchant marks his goods at ₹ 300 and allows a discount of 25%. If he still gains 12.5%, then the cost price of article is :
(a) ₹ 220 (b) ₹ 200
(c) ₹ 240 (d) ₹ 260

28 An item costing ₹ 200 is being sold at 10% loss. If the price is further reduces by 5%, the selling price will be :
(a) ₹ 170 (b) ₹ 171 (c) ₹ 180 (d) ₹ 181

29 A person sold two cows each for ₹ 9900. If he gained 10% on one and lost 20% on the other, then which of the following is true?
(a) He gained ₹ 200
(b) He lost ₹ 200
(c) He neither gained nor lost
(d) None of the above

30 Two third of a consignment was sold at a profit of 5% and the remainder at a loss of 2%. If the total profit was ₹ 400, the value of the consignment (in rupees) was :
(a) 15000 (b) 20000 (c) 10000 (d) 12000

31 A fruit seller declares that he sells fruits at the cost price. However, he uses a weight of 450 g instead of 500 g. His percentage profit is :
(a) 10% (b) $11\frac{1}{9}\%$ (c) 12% (d) $12\frac{2}{9}\%$

32 A person loses ₹ 20 by selling some bananas at the rate of ₹ 3 per banana and gains ₹ 30, if he sells them at ₹ 3.25 per banana. The number of bananas sold by him:
(a) 100 (b) 200 (c) 120 (d) 2400

33 Due to an increase of 30% in the price of eggs, 3 eggs less are available for ₹ 9.10. The present rate per egg is :
(a) 91 paise (b) 78 paise
(c) 48 paise (d) 84 paise

34 By selling 12 apples for a rupee, a man loses 20%. How many for a rupee should he sell to gain 20%?
(a) 8 (b) 10 (c) 15 (d) 16

35 A dealer buys a washing machine, listed at ₹ 10000 and gets 10% and 20% successive discounts. He spends 10% of his CP on transport. At what price (in rupees) should he sell the washing machine to earn a profit of 10%?
(a) 8722 (b) 7892
(c) 8712 (d) 8840

36 6% more is gained by selling a coat for ₹ 1425 than by selling it for ₹ 1353. The cost price of the coat is :
(a) ₹ 1000 (b) ₹ 1250
(c) ₹ 1500 (d) ₹ 1200

37 By selling a wrist watch at ₹ 405 the shopkeeper incurs a loss of 10%. What is the gain or loss percentage if he sells the same watch at ₹ 465?
(a) profit of 10% (b) loss of 6%
(c) profit of 3.33% (d) no profit no loss

38 Titan sells a wrist watch to a wholesaler making a profit of 10%. The wholesaler, in turn, sells it to the retailer making a profit of 10%. A customer purchases it by paying ₹ 990. Thus, the profit of retailer is $2\frac{3}{11}\%$. What is the cost incurred by the Titan to produce it?
(a) 768 (b) 750
(c) 800 (d) 820

39 Pepsi and Coke, there are two companies, selling the packs of cold-drinks. For the same selling price Pepsi gives two successive discounts of 10% and 25%. While Coke sells it by giving two successive discounts of 15% and 20%. What is the ratio of their marked price?
(a) 143 : 144 (b) 19 : 11
(c) 136 : 135 (d) 73 : 77

40 When a shopkeeper reduces the selling price from 1080 to 1026 its loss increases by 4 percentage point. What is the selling price of this same article when it fetches a profit of 4%?
(a) ₹ 1392 (b) ₹ 1404
(c) ₹ 1450 (d) ₹ 1350

41 The difference between CP and SP of a table fan is ₹ 175 when it gives the profit of 14%. What is the selling price of that fan?
(a) 1225 (b) 1450 (c) 1425 (d) 1275

42 A company instead of raising the mark-up by 20% discounted the cost price by 20% while stiching the price tag on its product. Further the company offers a discount of 6.25% to its customer. In this process company incurs a loss of ₹ 37.5 on a single article. What is the selling price of that article?
(a) 417.5 (b) 112.5
(c) 365.5 (d) none of these

43 When an article is sold for ₹ 703 loss incurred is 25% less than the profit earned on selling it at ₹ 836. What is the selling price of the article when it earns a profit of 20%?
(a) 912 (b) 1576
(c) 1532 (d) 1092

44 Arun bought toffees at 6 for a rupee. How many for a rupee he should sell to gain 20%?
(a) 3 (b) 4
(c) 5 (d) Can't be determined

45 A scientific calculator is available at Universal Shoppe in Hazratganz at 20% discount and the same is available at only 15% discount at Universal Shoppe Bhootnath Market. Ms. Aggarwal has just sufficient amount of ₹ 800 to purchase it at Universal Shoppe Hazratganz. What is the amount that Ms. Aggarwal has less than the required amount to purchase it at Universal Shoppe Bhootnath?
(a) ₹ 70
(b) ₹ 50
(c) ₹ 100
(d) data insufficient

46 A balance of a trader weighs 10% less than it should be. Still the trader marks-up his goods to get the overall profit of 20%. What is the markup on the cost price?
(a) 40% (b) 8%
(c) 25% (d) 16.66%

47 ITC sells one product at a profit of 20% another at a loss of 20% at the same selling price. What is the loss incurred by ITC?
(a) 1% (b) 2%
(c) 4% (d) 0%

48 In the previous question, if SP of each article be ₹ 200, what is the amount of loss?
(a) ₹ 10 (b) ₹ 16
(c) ₹ 16.66 (d) none of these

49 The cost price of 19 articles is same as the selling price of 29 articles. What is the loss %?
(a) 35% (b) 34.48%
(c) 52.63% (d) none of these

50 The selling price of 13 articles is same as the cost price of 23 articles. What is the profit percentage?
(a) 43.47% (b) 74.83%
(c) 78% (d) 76.92%

51 A trader can procure 34 pencils by selling 28 pencils. What is the ratio of cost price to the selling price of a pencil?
(a) 2 : 3 (b) 14 : 17
(c) 9 : 7 (d) 4 : 7

52 At style cloth emporium the shopkeeper measures 20% less for every metre of cloth also he marks-up goods by 20%. What is the profit percentage?
(a) 50% (b) 80%
(c) 75% (d) none of these

53 A bookseller procures 40 books for ₹ 3200 and sells them at a profit equal to the selling price of 8 books. What is the selling price of one dozen books, if the price of each book is same?
(a) 720
(b) 960
(c) 1200
(d) 1440

54 The profit percentage of A and B is same on selling the articles at ₹ 1800 each but A calculates his profit on the selling price while B calculates it correctly on the cost price which is equal to 20%. What is the difference in their profits?
(a) ₹ 360 (b) ₹ 60
(c) ₹ 540 (d) ₹ 450

55 Each of A and B sold their article at ₹ 1818 but A incurred a loss of 10% while B gained by 1%. What is the ratio of cost price of the articles of A to that of B?
(a) 101 : 90 (b) 85 : 89
(c) 81 : 75 (d) none of these

56 A trader sold an article at a loss of 5% but when he increased the selling price by ₹ 65 he gained 3.33% on the cost price. If he sells the same article at ₹ 936, what is the profit percentage?
(a) 15% (b) 16.66%
(c) 20% (d) data insufficient

57 Even after a discount of q% on marked price a trader gains by p%. What is the markup percentage over the cost price?
(a) $\frac{p+q}{(q-p)} \times 100$ (b) $\frac{p+q}{(100-p)} \times 100$
(c) $\frac{p+q}{(100-q)} \times 100$ (d) not possible

58 A milkman mixes 10% water in pure milk but he is not content with it so, he again mixes 10% more water in the previous mixture. What is the profit percentage of milkman if he sells it at cost price :
(a) 11.11% (b) 20%
(c) 21% (d) 12.1%

59 A person sold an electronic watch at ₹ 96 in such a way that his percentage profit is same as the cost price of the watch. If he sells it at twice the percentage profit of its previous percentage profit, then the new selling price will be :
(a) ₹ 132 (b) ₹ 150
(c) ₹ 192 (d) ₹ 180

60 A trader mixes 25% kerosene to his petrol and then he sells the whole mixture at the price of petrol. If the cost price of kerosene be 50% of the cost price of petrol, what is the net profit percentage?
(a) $11\frac{1}{9}\%$ (b) $12\frac{1}{9}\%$ (c) $9\frac{1}{11}\%$ (d) 20%

61 A retailer cheats both to his whole-seller and his customer by 10% by his faulty balance *i.e.,* he actually weighs 10% more while purchasing from wholesaler and weighs 10% less while selling to his customer. What is his net profit percentage, when he sells at CP?
(a) $22\frac{2}{11}\%$ (b) $22\frac{2}{9}\%$
(c) 20% (d) 21%

62 A trader procures his goods from a wholesaler, whose balance reads 1200 g for 1000 g. The trader sells all the procured goods to a customer after marking up the goods at 20% above the cost price. What is his overall percentage profit or loss in the whole transaction?
(a) 38% profit
(b) 50% profit
(c) no profit no loss
(d) none of the above

63 A person wants to reduce the trade tax, so he calculates his profit on the sale price instead of on the cost price. In this way by selling a article for ₹ 280 he calculates his profit as $14\frac{2}{7}\%$. What is his actual profit percentage?
(a) 20% (b) 16.66%
(c) 25% (d) data insufficient

64 A vendor sells his articles at a certain profit percentage. If he sells his articles at 1/3 of his actual selling price, then he incurs a loss of 40%. What is his actual profit percentage?
(a) 72% (b) 120%
(c) 80% (d) none of these

65 A retailer increases the selling price by 25% due to which his profit percentage increases from 20% to 25%. What is the percentage increase in cost price?
(a) 20% (b) 30%
(c) 25% (d) 50%

66 Abhinav saves ₹ 25 by getting 6.66% discount on a textbook. What is the amount of money (in ₹) paid by him?
(a) 450 (b) 350
(c) 225 (d) 375

67 At kul-kul petrol pump the operator gives 5% less petrol but he sells it at the cost price. What is his profit in this way?
(a) 5% (b) 5.6%
(c) 5.26% (d) 4.78%

68 Due to reduction of 25% in price of oranges a customer can purchase 4 oranges more for ₹ 16. What is original price of an orange?
(a) Re 1 (b) ₹ 1.33
(c) ₹ 1.5 (d) ₹ 1.6

69 A reduction of 20% in the price of sugar enables a housewife to purchase 6 kg more for ₹ 240. What is the original price per kg of sugar?
(a) ₹ 10 per kg (b) ₹ 8 per kg
(c) ₹ 6 per kg (d) ₹ 5 per kg

70 A wholesaler sells toys at a profit of 20% to a retailer and retailer sells these toys to its customer at a profit of 25%. What is the profit percentage of the retailer?
(a) 5% (b) 80%
(c) 20% (d) 25%

40 An egg seller sells his eggs only in the packs of 3 eggs, 6 eggs, 9 eggs, 12 eggs etc., but the rate is not necessarily uniform. One day Raju (which is not the same egg seller) purchased at the rate of 3 eggs for a rupee and the next hour he purchased equal number of eggs at the rate of 6 eggs for a rupee. Next day he sold all the eggs at the rate of 9 eggs for ₹ 2. What is his percentage profit or loss?
(a) 10% loss (b) 11.11% loss
(c) 3% loss (d) 2.5% profit

41 Virendra and Gurindra purchased one camera each at the same prices. Later on Amrendra purchased both cameras at equal prices from Virendra and Gurindra. But the profit percentage of Virendra was P while the same of Gurindra was Q, since Gurindra calculated his profit on the selling price. Thus $Q = 41\frac{2}{3}\%$ of P. If Amrendra sells one of the camera to Dholakiya at P% profit, then what is the cost price for Dholakiya, while Amrendra purchased each of the camera at ₹ 240?
(a) ₹ 676 (b) ₹ 500 (c) ₹ 576 (d) none

42 A merchant earns 25% profit in general. Once his 25% consignment was abducted forever by some goondas. Trying to compensate his loss he sold the rest amount by increasing his selling price by 20%. What is the new percentage profit or loss?
(a) 10% loss (b) 12.5% loss
(c) 12.5% profit (d) 11.11% loss

43 A milkman purchases 10 litres of milk at ₹ 7 per litre and forms a mixture by adding freely available water which constitutes 16.66% of the mixture. Later on he replaced the mixture by some freely available water and thus the ratio of milk is to water is 2 : 1. He then sold the new mixture at cost price of milk and replaced amount of mixture at twice the cost of milk, then what is the profit percentage?
(a) 68%
(b) 34%
(c) 40%
(d) none of the above

44 Profit on selling 10 candles equals selling price of 3 bulbs. While loss on selling 10 bulbs equals selling price of 4 candles. Also profit percentage equals to the loss percentage and cost of a candle is half of the cost of a bulb. What is the ratio of selling price of candle to the selling price of a bulb?
(a) 5 : 4 (b) 3 : 2
(c) 4 : 5 (d) 3 : 4

45 Cost price of two motorcycles is same. One is sold at a profit of 15% and the other for ₹ 4800 more than the first. If the net profit is 20%. Find the cost price of each motorcycle :
(a) ₹ 48000 (b) ₹ 52000
(c) ₹ 36000 (d) ₹ 42500

Answers

✓ *Introductory Exercise 6.1*

1 ₹ 180 and ₹ 225 2 ₹ 9000 3 ₹ 345 4 ₹ $153\frac{1}{3}$ per hundred 5 ₹ 15300, ₹ 17850, gain = $\frac{5}{11}\%$

✓ *Introductory Exercise 6.2*

1 *(d)*	2 *(c)*	3 *(d)*	4 *(b)*	5 *(c)*	6 *(b)*	7 *(c)*	8 *(d)*	9 *(c)*	10 *(c)*
11 *(c)*	12 *(b)*	13 *(c)*	14 *(b)*	15 *(a)*	16 *(d)*	17 *(c)*	18 *(b)*	19 *(c)*	

✓ **Level 01** *Basic Level Exercise*

1 *(a)*	2 *(b)*	3 *(b)*	4 *(c)*	5 *(c)*	6 *(d)*	7 *(d)*	8 *(b)*	9 *(a)*	10 *(c)*
11 *(a)*	12 *(a)*	13 *(d)*	14 *(d)*	15 *(c)*	16 *(d)*	17 *(b)*	18 *(b)*	19 *(c)*	20 *(a)*
21 *(d)*	22 *(c)*	23 *(c)*	24 *(b)*	25 *(c)*	26 *(a)*	27 *(b)*	28 *(b)*	29 *(d)*	30 *(a)*
31 *(b)*	32 *(b)*	33 *(a)*	34 *(a)*	35 *(c)*	36 *(d)*	37 *(c)*	38 *(c)*	39 *(c)*	40 *(b)*
41 *(c)*	42 *(b)*	43 *(a)*	44 *(c)*	45 *(b)*	46 *(b)*	47 *(c)*	48 *(c)*	49 *(b)*	50 *(d)*
51 *(b)*	52 *(a)*	53 *(c)*	54 *(b)*	55 *(a)*	56 *(c)*	57 *(c)*	58 *(c)*	59 *(a)*	60 *(a)*
61 *(b)*	62 *(c)*	63 *(b)*	64 *(c)*	65 *(a)*	66 *(b)*	67 *(c)*	68 *(b)*	69 *(a)*	70 *(d)*

✓ **Level 02** *Higher Level Exercise*

1 *(b)*	2 *(c)*	3 *(c)*	4 *(a)*	5 *(d)*	6 *(d)*	7 *(a)*	8 *(a)*	9 *(c)*	10 *(a)*
11 *(c)*	12 *(c)*	13 *(b)*	14 *(c)*	15 *(c)*	16 *(d)*	17 *(a)*	18 *(b)*	19 *(b)*	20 *(b)*
21 *(d)*	22 *(a)*	23 *(b)*	24 *(b)*	25 *(a)*	26 *(b)*	27 *(a)*	28 *(c)*	29 *(b)*	30 *(b)*
31 *(c)*	32 *(b)*	33 *(c)*	34 *(b)*	35 *(c)*	36 *(a)*	37 *(a)*	38 *(a)*	39 *(a)*	40 *(b)*
41 *(c)*	42 *(c)*	43 *(a)*	44 *(b)*	45 *(a)*					

Hints & Solutions

☑ Level 01 *Basic Level Exercise*

1

	Abhishek	Bhanu
Discount	2800	2400 + 640 = 3040

The difference in selling price is same as difference in discount which is ₹ 240 = (3040 – 2800)

2 ∵ 10 % of x = 15% of y, where $x + y = 30000$

$$\Rightarrow \quad \frac{x}{y} = \frac{3k}{2k}$$

Hence, the difference = k = 6000

3 Let the cost price of 1 litre pure milk be ₹ 1, then

$$\left.\begin{matrix}6 \text{ litres (milk)} \rightarrow \text{CP} = ₹\,6 \\ 2 \text{ litres (water)} \rightarrow \text{CP} = ₹\,0\end{matrix}\right\} \rightarrow \text{CP} = ₹\,6 \text{ only.}$$

and 8 litre mixture → SP → 8 × 2 = ₹ 16

$$\text{Profit} = \frac{16-6}{6} \times 100 = \frac{1000}{6} = 166.66\%$$

4 $\left.\begin{matrix}\text{SP of } 60\% \text{ goods} = 0.6x \times 0.95 = 0.57x \\ \text{SP of } 40\% \text{ goods} = 0.4x \times 1.1 = 0.44x\end{matrix}\right\}$ Total SP = $1.01x$

$$\text{Profit} = 0.01x = 100$$

$$\Rightarrow \quad x = 10000$$

Alternatively From option (c)

10000 → 6000, 4000

6000 → loss = 300; 4000 → gain = 400

net gain = 400 – 300 = ₹ 100,

Hence, option (c) is correct.

5 Let the MP of 1 kg tea be ₹ 1, then CP of 20 kg with discount = 20 × 0.9 = ₹ 18

Also 1 kg tea is free. So, the retailer gets tea worth ₹ 21 by paying ₹ 18 only.

$$\text{Profit \%} = \frac{\text{goods left}}{\text{goods sold}} \times 100$$

$$= \frac{21-18}{18} \times 100 = 16.66\%$$

(Since, the retailer earns ₹ 3 on each ₹ 18)

6 Let the CP of profit yielding article be ₹ 100, then

$$\left.\begin{matrix}250 \quad \text{CP} \\ \downarrow \\ 175 \quad \text{SP}\end{matrix}\right\} 30\% \text{ loss}$$

$$75\% \text{ profit} \left\{\begin{matrix}\text{SP} \quad 175 \quad \leftrightarrow \quad 175 \quad \text{SP} \\ \uparrow \\ \text{CP} \quad 100\end{matrix}\right.$$

Total CP = 350, Total SP = 350

So, there is no profit no gain.

Alternatively $1.75x = 0.7y$

$$\Rightarrow \quad \frac{x}{y} = \frac{2}{5}$$

Again $\left[2 \times 1.75 = 3.5\right]$

and $\left[5 \times 0.7 = 3.5\right]$

Total cost price = 2 + 5 = ₹ 7

Total selling price = 3.5 + 3.5 = ₹ 7

Hence, no loss no gain.

7

	CP	SP	MP
Initially :	100	140	× (since, profit = 40%)
New prices :	100	112	140
		profit = 12%	discount = 20%

8 $\frac{\text{CP}}{\text{SP}} = \frac{2}{3}$

$$\therefore \quad \text{profit (\%)} = \frac{1}{2} \times 100 = 50\%$$

Alternatively $\text{CP} = \frac{100}{18} = 5.55$ paise

$$\text{SP} = \frac{100}{12} = 8.33 \text{ paise}$$

$$\text{Profit} = \frac{8.33 - 5.55}{5.55} \times 100 = 50\%$$

9

CP	SP	MP
100	120	150
	20% profit	20% discount

But it can be directly solved as SP = 1.2 CP = ₹ 2304

NOTE There is no role of discount.

10 **NOTE** There is no role of cost price of article (₹ 123684)

(Consider) CP → 100

16.66% of profit + 60, 40 → 10, –24 → 60% loss

70, 16

(100 –14) = SP 86

Since, the overall % loss = 14%.

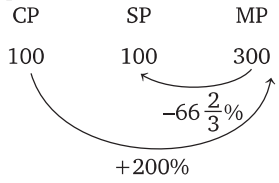
Thus, option (c) is correct.

Alternatively $0.6x + k \times 0.4x = 0.86x$

$$\Rightarrow \quad 0.4kx = 0.16x \Rightarrow k = 0.4$$

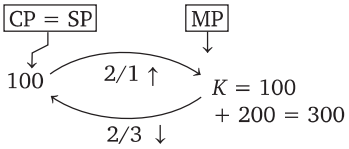
Therefore, loss = 1 – 0.4 = 0.6 *i.e.,* 60%

11 Go through options



NOTE When there is no loss, CP = SP.

Alternatively



From percentage change graphic :

Decrease	Increase
$\frac{2}{3}\downarrow$	$\frac{2}{3-2}=\frac{2}{1}\uparrow$
= 66.66%	= 200%

So, when CP = 100, markup % = 200, MP = 300

$$\text{Discount} = 66\frac{2}{3}\% = 200,\ \text{SP} = 100 = \text{CP}$$

Alternatively $(\text{CP} = \text{SP}) = \frac{1}{3}\text{MP}$

$$\Rightarrow \quad (\text{CP} = \text{SP}) : \text{MP} = 1 : 3$$

12 $$\text{CP} : \text{MP} = 2x : 3x$$

$$\Rightarrow \quad \text{Profit} = x$$

(%) profit : (%) discount = 3 : 2

Let CP = 200, SP = 300

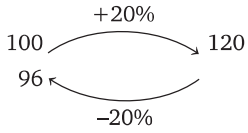
But $$\frac{3x}{100} \times 200 + \frac{2x}{100} \times 300 = 100$$

$$\Rightarrow \quad x = 8.33\%$$

$$\text{Discount } 2x = 16.66\%$$

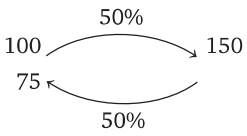
13 It is dependent upon the markup (%) or discount (%)

Since,



loss = 4%

Again,



loss = 25%

Hence, we can't determine.

14 CP → Profit (%) → MP

$$100 \to x\% \to 100 + \frac{100x}{100} = (100 + x)$$

Again, MP → discount → SP

$$(100 + x) \to x\% \to (100 + x) - (100 + x)\frac{x}{100}$$

$$= (100 + x)\left(1 - \frac{x}{100}\right)$$

Now, $$\frac{(100 + x)}{(100 + x)\left(1 - \frac{x}{100}\right)} = \frac{2}{1}$$

$$\Rightarrow \quad \left(1 - \frac{x}{100}\right) = \frac{1}{2} \quad \Rightarrow \quad x = 50\%$$

This can be observed in the solution of the previous problem.

15 Go through option

$$180 \times 12 \times 1.2 + 180 \times 8 \times 1.1 = 180\,[14.4 + 8.8]$$

$$= 180\ (23.2) = 4176$$

and $$180 \times 20 \times 1.15 = 4140$$

Therefore, loss = 4176 − 4140 = 36

Hence, option (c) is correct.

16 Gain = 30000

loss = 20000

Net gain = 10000 over ₹ 4 lakh

Hence, $$\text{profit} = \frac{10000}{400000} \times 100 = 2.5\%$$

17 Let the MP = ₹ 1 per kg, then

Weight	MP	Rate
100	100	1
96	$\frac{80}{\text{SP}}$	$\frac{80}{96}$

$$\text{Effective discount} = 1 - \frac{80}{96} = \frac{16}{96}$$

$$\%\ \text{discount} = \frac{16}{96} \times 100 = 16.66\%$$

18 CP of $A + B + C = 2x \times y + 5x \times 2y + 2x \times 4y = 20xy$

Profit of $A = 0.2\,x\,y$

Profit of $B = 2\,x\,y$

Profit of $C = 2\,x\,y$

Total profit $= 4.2\,x\,y$

$$\text{Profit \%} = \frac{4.2xy}{20xy} \times 100 = 21\%$$

19 Initially

	CP	Profit	SP	MP
	100	x	$(100 + x)$	133.33
After change	100	$2x$	$(100 + x)\frac{7}{6}$	

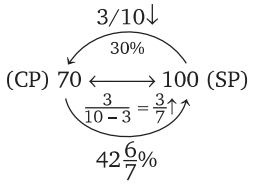
Now, since $(100 + x)\frac{7}{6} - 100 = 2x$

$$\Rightarrow \quad x = 20\%$$

	CP	Profit	SP	MP
	100	20	120	133.33
So,	300	60	360	400
Again,	300	120	420	

So, the increased selling price = ₹ 420

20 From percentage change graphic



$$\text{Decrease} = \frac{3}{10}\downarrow = 30\%$$

$$\text{Increase} = \frac{3}{(10-3)} = \frac{3}{7}\uparrow = 42\frac{6}{7}\%$$

21 Let the price be x, then

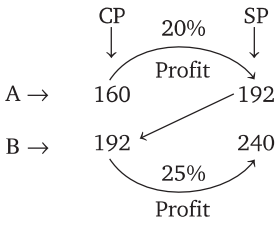
$$x \times 0.8 \times 0.9 = 468$$

$\Rightarrow x = 650$, therefore marked price = ₹ 650

Alternatively $650 \times 0.8 \times 0.2 = 468$

Hence, option (d) is correct.

22



23 Reduced price $= 100 \times 0.95 \times 0.90 \times 0.80 = 68.40$

$\therefore$ Single discount $= (100 - 68.4)\% = 31.6\%$

24 $\text{CP} = \frac{10}{12}$, $\text{SP} = \frac{12}{10}$

$$\text{Profit (\%)} = \frac{\frac{12}{10} - \frac{10}{12}}{\frac{10}{12}} \times 100 = 44\%$$

25 Let the cost price of one gram be ₹ 1, then the markup price be ₹ 1.2 per g. Now, he sells 1000 g which seems to be 1250 g so he charges to a customer $1250 \times 1.2 =$ ₹ 1500 for 1000 g (or ₹ 1000)

Thus, his profit % $= \frac{1500 - 1000}{1000} \times 100 = 50\%$

26 Let the CP be ₹ x, then SP be $0.96x$

$\therefore \quad 0.96x = 240 \Rightarrow x = 250$

Now, the new SP $= 250 \times 1.1 = 275$

27

CP	SP	MP
x	$112.5x = 225$	300
		(−25%)

$\therefore x = 200$, hence the cost price be ₹ 200.

NOTE It can also be solved by using option.
First of all find the SP by decreasing MP by 25%, then this SP will be equal to 112.5% (12.5% is the profit) of the cost price, so the CP can be find as given above.

28

CP	$\xrightarrow{(-10\%)}$	$(SP)_1$	$\xrightarrow{(-5\%)}$	$(SP)_2$
200		180		171

29 The CP of profitable cow $= \frac{9900}{1.1} = 9000$

and profit = ₹ 900

The CP of loss yielding cow $= \frac{9900}{0.8} = 12375$

and loss = ₹ 2475

So, the net loss $= 2475 - 900 = 1575$

30 The best way is to go through option. Consider option (a)

15000

$\left(\frac{2}{3}\right)$ $\left(\frac{1}{3}\right)$

10000 5000

(5% profit) ↓ ↓ 2% loss

500 100

Net gain = ₹ 400 (= 500 − 100)

Hence, option (a) is correct.

Alternatively $\left[\left(1.05 \times \frac{2x}{3} + 0.98 \times \frac{x}{3}\right) - x\right] = 400$

$\Rightarrow \quad x = 15000$

31 Profit (%) $= \frac{\text{goods left}}{\text{goods sold}} \times 100 = \frac{50}{450} \times 100 = 11\frac{1}{9}\%$

Alternatively Suppose CP of 1 g is ₹ 1, then he sells goods worth ₹ 450 only and charges ₹ 500. So, the profit is ₹ 50 over the sale price of ₹ 450.

32 Go through options. Consider option (b)

$200 \times 3.25 - 200 \times 3 =$ ₹ 50

$= (30 + 20)$

Hence, option (b) is correct.

Alternatively ₹ 0.25 is the difference on 1 banana

$\therefore$ ₹ 1 is the difference on $\frac{1}{0.25} = 4$ bananas

$\therefore$ ₹ 50 is the difference on $4 \times 50 = 200$ bananas

33 By percentage change (or fraction change) graphic

increase	decrease	
$30\% = \frac{3}{10}\uparrow$	$\frac{3}{13}\downarrow$	[since, product *i.e.*, price is constant]

It means now, $\frac{3}{13}$ times less eggs are available which is equal to 3 eggs.

i.e., $\frac{3x}{13} = 3$

$\Rightarrow \quad x = 13$ eggs (initially)

Now, available eggs on the same price $= 13 - 3 = 10$

Thus, the new price $= \frac{910}{10} = 91$ paise

Alternatively Go through options.

	New or changed	No. of eggs		Rate = total prices
Initial →	(+3)	10	× 91	= 910 paise
		13	× 70	= 910 paise (+30%)

Therefore, option (a) is correct.

34 $\text{SP} = \frac{100}{12}$ paise

$\text{CP} = x$, $\text{SP} = 0.8x = \frac{100}{12}$

$\Rightarrow \quad x = \frac{100}{9.6}$ paise

Therefore, $\text{CP} = \frac{100}{9.6}$ paise

Thus, the new SP (with 20% profit)

$= \frac{100}{9.6} \times 1.2 = \frac{100}{8}$ paise

Hence, 8 apples can be purchased for ₹ 1 to gain 20%.

35 $10000 \xrightarrow{-10\%} 9000 \xrightarrow{-20\%} 7200 \xrightarrow{+10\%} 7920 \xrightarrow{+10\%} 8712$

36 $1425 - 1353 = 72 = 6\%$ of CP

$\Rightarrow \quad \text{CP} = 1200$

37 $\text{CP} = \frac{405}{0.9} = 450$

New SP = ₹ 465

$\text{Profit (\%)} = \frac{465 - 450}{450} \times 100 = 3.33\%$

38 $\left[((x \times 1.1) \times 1.1) \times \frac{1125}{1100}\right] = 990 \Rightarrow x = 800$

Alternatively Go through options or by the reverse process.

39 Best way is to go through option

Alternatively $P \times 0.9 \times 0.75 = C \times 0.85 \times 0.80$

$\Rightarrow \quad \frac{P}{C} = \frac{136}{135}$

40 Let the CP be x, then

$0.04x = 54 = (1080 - 1026)$

$\Rightarrow \quad x = \frac{54}{0.04} = 1350$ (CP)

$\text{CP} = 1350$

$\text{SP} = 1350 \times 1.04 = 1404$

41 $\text{SP} - \text{CP} = 175 = 14\%$ of CP

$\Rightarrow \quad \text{CP} = 1250$

$\therefore \quad \text{SP} = 1250 + 175 = 1425$

42 CP = 100, then

Tag price = 80

SP, after discount = 75

Total loss = 25

∵ ₹ 25 are being less at CP of ₹ 100

∴ ₹ 37.5 are being less at CP of ₹ $\frac{100}{25} \times 37.5 = 150$

Hence, $\text{SP} = 150 - 37.5 = 112.5$

Alternatively $112.5 + 37.5 = 150 = \text{CP}$

Again, $150 \times 0.8 \times .9375 = 112.5 = \text{SP}$

Hence, option (b) is correct.

43 $\text{CP} + 4k = \text{SP}_1$ (gives profit)

$\text{CP} - 3k = \text{SP}_2$ (gives loss)

Since, loss ($3k$) is 25% less than profit ($4k$)

$\therefore \quad \text{SP}_1 - \text{SP}_2 = 7k = 836 - 703 = 133 \Rightarrow k = 19$

$\therefore \quad \text{CP} = \text{SP}_1 - 4k = \text{SP}_2 + 3k = 760$

Therefore, required $\text{SP} = 760 \times 1.2 = 912$

44 You can go through options.

$\text{CP} = \frac{100}{6} = 16.66$ paise

$\text{SP} = \frac{100}{5} = 20$ paise

$\text{Profit (\%)} = \frac{20 - 16.66}{16.66} \times 100 = \frac{3.33}{16.66} \times 100$

$= \frac{1}{5} \times 100 = 20\%$

Hence, option (c) is correct.

Alternatively $\text{CP} = \frac{100}{6}$, $\text{SP} = \frac{100}{6} \times 1.2 = \frac{100}{5}$

Hence, he should sell 5 toffees for ₹ 1 (= 100 paise)

45

CP	MP
$800 = 0.8x$	x

20% discount

$\Rightarrow \quad x =$ ₹ 1000 = MP

SP at Hazratganz = ₹ 800

SP at Bhootnath $= 1000 \times 0.85 = 850$

So, she requires ₹ 50 more to purchase the same calculator at Bhootnath.

46 Let the CP be ₹ 1 per g

But he weighs 900 g for every 1000 g

∴ Value of goods sold = 900

Now, let the markup be x%

$\therefore \quad \text{MP} = 1000 + \frac{1000x}{100} = (1000 + 10x)$

But, since MP = SP, ∴ $\text{SP} = (1000 + 10x)$

Hence, profit (%) $= \frac{(1000 + 10x) - 900}{900} \times 100 = 20 \Rightarrow x = 8$

Thus, the markup = 8%.

47 $\text{Loss \%} = \left(\frac{\text{common gain or loss}}{10}\right)^2 \% = \left(\frac{20}{10}\right)^2 \% = 4\%$

48 Since, loss = 4%

Now, assume total CP of both articles be x, then

$\text{SP} = 0.96x = 400 \Rightarrow x = \frac{400}{0.96} = \text{CP}$

$\therefore \quad$ loss = 4% of CP

$\frac{4}{100} \times \frac{400}{0.96} =$ ₹ 16.66

49 $$\frac{CP}{SP}=\frac{29x}{19x}$$

$$\text{Loss \%}=\frac{29x-19x}{29x}\times 100=34.48\%$$

(To understand the concept assume CP of each article ₹ 29 and SP of each article = ₹ 19)

50 $\frac{CP}{SP}=\frac{13}{23}$. Profit (%) $=\frac{23-13}{13}\times 100=\frac{10}{13}\times 100=76.92\%$

51 $\frac{CP}{SP}=\frac{28}{34}=\frac{14}{17}$

52 MP = ₹ 1.2, if CP = ₹ 1

But he gives the cloth worth ₹ 0.8 instead of ₹ 1.

Therefore, profit % $=\frac{1.2-0.8}{0.8}=\frac{0.4}{0.8}\times 100=50\%$

53 CP = ₹ 80 $=\left(\frac{3200}{40}\right)$

Now, SP of 40 books = CP of 40 books + SP of 8 books

$\Rightarrow$ SP of 32 books = 3200

$\Rightarrow$ SP of 1 book = ₹ 100

$\therefore$ Required SP of 1 dozen books = ₹ 1200

54 Profit of $A=1800\times\frac{20}{100}=$ ₹ 360

$$\text{CP of } B=\frac{1800}{1.2}=1500$$

$\therefore$ Profit of B = ₹ 300 = (1800 – 1500)

$\therefore$ Difference in profit of A and B = ₹ 60

55 $$\text{CP of } A=\frac{1818}{0.9}=2020$$

$$\text{CP of } B=\frac{1818}{1.01}=1800$$

$$\frac{\text{CP of } A}{\text{CP of } B}=\frac{2020}{1800}=\frac{101}{90}$$

56 103.33 CP – 0.95 CP = 65 $\Rightarrow$ CP = ₹ 780

$$\text{Profit (\%)}=\frac{936-780}{780}\times 100=20\%$$

57 Best way is to assume some proper values, then verify it.

58 Let us assume initially he has 100 l of milk

So, after 10% of water addition new amount of mixture = 110 l

Again adding 10% more, then the final amount of mixture = 121 l

$$\text{Profit \%}=\frac{21}{100}\times 100=21\%$$

59 $$SP=x+\frac{x\times x}{100}=96\%\Rightarrow x=60$$

New, $SP=60+\left(\frac{60\times 120}{100}\right)=$ ₹ 132

60 $K\rightarrow 1$ litre, price of $K\rightarrow 1$ $P\rightarrow 4$ litre, price of $P\rightarrow 2$

Price of total (new) mixture $=1\times 1+4\times 2=$ ₹ 9

Price of pure petrol of same quantity $=5\times 2=$ ₹ 10

Percentage profit $=\frac{1}{9}\times 100=11\frac{1}{9}\%$

61 Amount purchased = 1100 g

Amount sold = 900 g

$$\text{Profit \%}=\frac{200}{900}\times 100=22\frac{2}{9}\%$$

62 If CP be ₹ 1 per g, then he pays ₹ 1200 for 1000 g

Again, he obtains ₹ 1200 for 1000 g (by selling at 20% profit). Thus, there is no loss no gain.

63

CP	SP
240	280

$\left(-14\frac{2}{7}\%\right)$

Actual profit (%) $=\frac{40}{240}\times 100=16.66\%$

NOTE Here you can apply base change concept.

increase	decrease
16.66%	$14\frac{2}{7}\%$
$\frac{1}{6}\uparrow$	$\frac{1}{7}\downarrow$

64 Go through options : $100\xrightarrow{80\%}180$

$$\text{New SP}=\frac{180}{3}=60$$

Percentage loss = 40% (100 – 60)

Hence, (c) is correct choice.

65

CP	Profit	SP	
100	20	120	
x	25	150	+25%

$x\times 1.25=150\Rightarrow x=120$

$$\text{\% change}=\frac{120-100}{100}\times 100=20\%$$

66 6.66% of MP = 25 $\Rightarrow$ MP = 375

$\therefore$ SP = MP – 25 = 350

67 Profit % $=\frac{5}{95}\times 100=5.26\%$

68 Recall it is based on inverse proportion or product constancy concept.

Reduction in price	increase in amount
$25\%\left(\frac{1}{4}\right)\downarrow$	$\uparrow\left(\frac{1}{3}\right)$ 33.33% = 4 oranges

It means original number of oranges $=4\times 3=12$

$\therefore$ Original price of oranges $=\frac{16}{12}=$ ₹ 1.33

69

Reduction in price	increase in amount
$20\%\left(\frac{1}{5}\right)\downarrow$	$\frac{1}{4}\uparrow(25\%)=6$ kg

It means original amount of sugar needed $=6\times 4=24$ kg

$\therefore$ Original price of the sugar $=\frac{240}{24}=$ ₹ 10 per kg.

70 Just a sitter. Don't be bogged down. Its simple *i.e.,* 25%, nothing else, which is very obvious.